Finitism Audit of the Flavour Suite

which calculations are exact, and how the continuum is controlled

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Abstract

The programme admits no completed infinity: an exact claim must live in finite, finite-field, or cyclotomic arithmetic and be checked exhaustively, with the continuum entering only as a labelled degenerate idealisation. This note audits the flavour suite against that standard. Every load-bearing claim of the mass sector — the Koide identity, the circulant amplitude form, the quarter-turn amplitude $\sqrt{2}$, the $\pi/12$ boundary, and the Gauss-sum reality and $\pi/3$-quantisation behind the winding phase — is re-verified *float-free* in `validation/exact_core.py`: the Koide and circulant facts in the cyclotomic field $\mathbb{Q}(\omega, \sqrt{2})$, the boundary as an exact rational multiple of π, and the Gauss-sum facts as *integer* identities in the group ring $\mathbb{Z}[\mathbb{Z}/3 \times \mathbb{Z}/p]$, with no complex numbers at all. Three genuine float relapses are identified and re-done; the remaining continuum operations — comparison to measured masses, the λ-power logarithms, the arccos phase extraction, and the random-coefficient mixing and robustness runs — are data comparisons or sensitivity analyses, each labelled, on which no exact claim depends. The suite is $17 + 17 + 12 + 10$ checks; no exact claim rests on a continuum construct.

1 The discipline

An exact claim must be an identity in a finite arena: a finite set, a finite field, the integers, or a cyclotomic ring $\mathbb{Z}[\zeta_n]$. Floating point, logarithms, numerical trigonometry, pseudo-random sampling, and tolerance comparisons are continuum tools; they may appear only as *informed approximations* — a comparison of an exact object to measured data, or a sensitivity analysis — never as the verification of an exact statement. We sort every check of the flavour suite into [EXACT] (a finite/cyclotomic identity) and [APPROX] (a labelled continuum comparison).

2 The exact core (float-free)

The load-bearing claims are verified with zero continuum operations (`exact_core.py`, 17/17).

The two Gauss-sum facts deserve a word, since they replace the most continuum-flavoured step. The cubic overlap $b = \sum_{u \in U} \chi_3(u)\psi(u)$ is, before any evaluation, the formal integer combination $\sum_{r,s} C_{rs}\, \omega^r \zeta_p^s$ with $C_{rs} = \#\{u : \chi_3(u) = \omega^r,\ \psi\text{-exponent} = s\}$. Conjugation sends $(\omega, \zeta_p) \mapsto (\omega^{-1}, \zeta_p^{-1})$, i.e. $C_{rs} \mapsto C_{-r,-s}$. "b real" is then the *integer* statement $C_{rs} = C_{-r,-s}$, and "$\arg b \in \frac{\pi}{3}\mathbb{Z}$" is b^3 real, the same symmetry of the integer coefficient array of the convolution b^3. Both are checked exactly for $p \in \{17, 29, 41, 53\}$, $p \equiv 5 \pmod{12}$, with no float and no tolerance.

3 Relapses found and re-done

Three checks in the working scripts verified an *exact* claim through floating point. Each is re-done above; the working scripts retain the fast numerical version, now labelled as a re-illustration.

claim	arena	verification
$\sum_k \sqrt{m_k} = 3a$, $\sum_k m_k = 3a^2 + 6\lvert b\rvert^2$	$\mathbb{Q}(\omega)$, ω^3=1	symbolic, ω-cross terms cancel
$Q = \frac{1}{3} + \frac{2}{3}\rho^2$; r=$\sqrt{2} \Rightarrow Q$=$\frac{2}{3}$	$\mathbb{Q}(\omega, \sqrt{2})$	$\sqrt{2} = \zeta_8 + \zeta_8^{-1}$, exact
circulant eigenvalues $a + b\omega^k + \bar{b}\omega^{2k}$	$\mathbb{Z}[\omega]$ matrix	$F^{-1}MF$ diagonal, symbolic
$\sqrt{M}$ Hermitian ($P^2 = P^\dagger$)	$\mathbb{Z}[\omega]$ matrix	$M - M^\dagger = 0$, symbolic
$\delta_{\text{LO}} = \frac{3\pi}{4} - \frac{2\pi}{3} = \frac{\pi}{12}$	rational π (ζ_{24})	exact symbolic
$1 + \sqrt{2}\cos\frac{3\pi}{4} = 0$ (massless at boundary)	$\mathbb{Q}(\sqrt{2})$	exact algebraic
symmetric cubic overlap $b = \bar{b}$ (real)	$\mathbb{Z}[\mathbb{Z}/3 \times \mathbb{Z}/p]$	integer coeff. symmetry
drive-twisted $b^3 = \bar{b}^3$ ($\arg b \in \frac{\pi}{3}\mathbb{Z}$)	$\mathbb{Z}[\mathbb{Z}/3 \times \mathbb{Z}/p]$	integer convolution, exact

Table 1: The exact core. ω is a primitive cube root of unity (minimal polynomial $x^2 + x + 1$, no π, no float); the Gauss-sum facts are integer identities — the overlap is an element of the group ring $\mathbb{Z}[\mathbb{Z}/3 \times \mathbb{Z}/p]$ and reality / $\pi/3$-phase are symmetries of its integer coefficient array, computed with no complex numbers.

(R1) **Circulant eigenvalues and Hermiticity** (`m10.py`, sec. 1) used `numpy.linalg.eigvalsh` and `allclose`. Re-done as an exact symbolic diagonalisation $F^{-1}MF$ over $\mathbb{Z}[\omega]$ (Table 1).

(R2) **Reality of the symmetric cubic overlap** (`delta.py`) used `cmath.exp` and a tolerance $\lvert\text{Im}\rvert < 10^{-8}$. Re-done as the integer identity $C_{rs} = C_{-r,-s}$.

(R3) **$\pi/3$-quantisation of the drive-twisted overlap** (`delta.py`) used a float "off-grid" tolerance $< 10^{-6}$. Re-done as the integer identity $b^3 = \bar{b}^3$.

A fourth item is a *representation* purification rather than a relapse: the Koide identity was already exact in `tier_b.py`, but written with cos and π; it is recast in the roots of unity ω and $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$, making the finite arena $\mathbb{Q}(\omega, \sqrt{2})$ explicit.

4 Controlled continuum approximations

The remaining continuum operations are all [APPROX]: comparisons of exact objects to data, or sensitivity analyses. None verifies an exact claim.

- **Comparison to measured masses** ($Q = 0.666661$, ρ, the $44.9997°$ angle, the per-sector r and δ): floating evaluation of cyclotomic objects against PDG inputs. The *identities* are exact; only the confrontation with data is numerical.
- **The λ-power texture** (`numpy.log`): the effective powers $\log(m_i/m_3)/\log\lambda$ read mass ratios against integer powers of λ. The exact content — role depths $(0, 1, 2)$ as flavour charges — is structural; the logarithm is a read-out chart.
- **The Gatto and Georgi–Jarlskog ratios** ($\sqrt{m_d/m_s}$, $(m_\mu/m_e)/(m_s/m_d)$): exact algebraic predictions evaluated on data.
- **The phase extraction** (arccos): a continuum reading of δ from measured masses, reported, with the structural value $\pi/12$ and the lock $1{:}\frac{1}{2}{:}\frac{1}{3}$ the actual content.
- **Random-coefficient runs** (RNG): the cross-sector robustness over PDG mass ranges and the lopsided/seesaw mixing demonstration are sensitivity analyses over experimental uncertainty and over $O(1)$ coefficients — explicitly illustrative, never a claim.

5 Verdict

No exact claim of the flavour sector depends on a continuum construct. The Koide value, the circulant form, the quarter-turn amplitude, the $\pi/12$ boundary, and the reality and $\pi/3$-quantisation behind the winding phase are identities in $\mathbb{Q}(\omega, \sqrt{2})$, in rational multiples of π (equivalently ζ_{24}), and in the integer group ring $\mathbb{Z}[\mathbb{Z}/3 \times \mathbb{Z}/p]$. The continuum enters only where the rest of the programme admits it — as the chart on which the finite structure is compared to measurement.